Strings

Characters and Strings

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Done

Content

- Introduction to characters and strings
- String conversion functions
- String manipulation functions

Introduction

- Strings are used for storing text/characters.
- For example, "Hello World" is a string of characters.
- Unlike many other programming languages, C does not have a String type to easily create string variables.
- Instead, you must use the char type and create an array of characters to make a string in C:

```c
char greetings[] = "Hello World!";
```

Another Way Of Creating Strings

- In the previous examples, we used a "string literal" to create a string variable.
- This is the easiest way to create a string in C.
- You should also note that you can create a string with a set of characters. This example will produce the same result as the previous example .

```
char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '!', '\0'};
printf("%s", greetings);
```

- The \0 character at the end is known as the "null terminating character", and must be included when creating strings using this method.

\0 \0 \0

\0

Access Strings

- Since strings are actually arrays in C, you can access a string by referring to its index number inside square brackets [ ].
- This example prints the first character (0) in greetings:

```
char greetings[] = "Hello World!";
printf("%c", greetings[0]);
```

strcmp

Loop Through a String

- You can also loop through the characters of a string, using a `for` loop statement

```
char carName[] = "Volvo";
int i;

for (i = 0; i < 5; ++i) {
  printf("%c\n", carName[i]);
}
```

sizeof() function

- Used to compute the size of its operand.

c-1
int-4
float-4
double-8
pointer=8

```c
// C Program To demonstrate
// sizeof operator
#include <stdio.h>
int main(){
    printf("Size of char: %lu bytes\n", sizeof(char));
    printf("Size of int: %lu bytes\n", sizeof(int));
    printf("Size of float: %lu bytes\n", sizeof(float));
    printf("Size of double: %lu bytes\n", sizeof(double));
    printf("Size of pointer: %lu bytes\n", sizeof(void *));
    return 0;
}
```

```
Size of char: 1 bytes
Size of int: 4 bytes
Size of float: 4 bytes
Size of double: 8 bytes
Size of pointer: 8 bytes
```

```c
#include <stdio.h>

int main() {
  char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l',
'd', '!', '\0'};
  char greetings2[] = "Hello World!";

  printf("%lu\n", sizeof(greetings));
  printf("%lu\n", sizeof(greetings2));

  return 0;
}
```

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```
13
13
```

char 1
int float 4
double 8

Cont..

String (\n)

```c
#include <stdio.h>
int main()
{
    int arr1[] = {1, 2, 3, 7, 98, 12, 35, 99}; // each value = 4 bytes
    printf("sizeof arr1 is :%lu ",sizeof(arr1));

    // each character = 1 byte
    char arr2[] = {'1', '2', '3', '4', '5', '6', '7', '8', '9','\0' };
    printf("\nsizeof arr2 is :%lu ",sizeof(arr2));

    char arr3[] = "1234798012359914"; // each char = 1 byte
    printf("\nsizeof arr3 is :%lu ",sizeof(arr3));

    double arr4[] = {12.124,12.3,15.4};    // each value = 8 bytes
    printf("\nsizeof arr4 is :%lu ",sizeof(arr4));

    printf("\nsizeof arr4[1] is :%lu ",sizeof(arr4[1]));

    return 0;
}
```

32

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\n

Remember

```
Size of char: 1 bytes
Size of int: 4 bytes
Size of float: 4 bytes
Size of double: 8 bytes
Size of pointer: 8 bytes
```

Output

```
sizeof arr1 is :32
sizeof arr2 is :10
sizeof arr3 is :17
sizeof arr4 is :24
sizeof arr4[1] is :8
```

Example of string

```c
char message[] = "Good to see you,";
char fname[] = "John";

printf("%s %s!", message, fname);
```

```
Good to see you, ~~John!~~
```

Str

len

Escape character

- use the backslash escape character. To turns special characters into string characters.

Escape character	Result	Description
\'	'	Single quote
\"	"	Double quote
\\	\	Backslash

- How to print "Al-Bukhary International Universiti"

Example

```
It's alright.
```

```c
char txt[] = "It\'s alright.";
```

```
The character \ is called backslash.
```

```c
char txt[] = "The character \\ is called backslash.";
```

```c
char txt[] = "First month is \"January\"";
printf("%s", txt);
```

First month is "January"

Others Escape character

Escape Character	Result
\n	New Line
\t	Tab
\0	Null

\n → New line
\t → Tab
\0 → Null

Statement	Output
`char txt[] = "Hello\nWorld!";` `printf("%s", txt);`	Hello World!
`char txt[] = "Hello\tWorld!";` `printf("%s", txt);`	Hello World!
`char txt[] = {'H', 'e', 'l', 'l', 'o', '\0'};` `printf("%s", txt);`	Hello

String Functions

1) String Length
2) Concatenate Strings
3) Copy Strings
4) Compare Strings
5) Others – visit [C string (string.h) Library Reference (w3schools.com)](https://www.w3schools.com/c/c_ref_string.php)
   https://www.w3schools.com/c/c_ref_string.php

len
cat
cpy
cmp

len
cat
cpy
cmp

strlen
strcat
strcpy
strcmp

cat
cpy
cmp

len
cat
cpy
cmp

String Length

- to get the length of a string, you can use the strlen() function:

```
#include <stdio.h>
#include <string.h>

int main() {
  char alphabet[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
  printf("%d", strlen(alphabet));
  return 0;
}
```

Strlen(alphabet) — 26

27

OUTPUT

26

Concatenate Strings

- To concatenate (combine) two strings, you can use the strcat() function:

```c
#include <stdio.h>
#include <string.h>

int main() {
  char str1[20] = "Hello ";
  char str2[] = "World!";

  // Concatenate str2 to str1 (the result is stored in str1)
  strcat(str1, str2);

  // Print str1
  printf("%s", str1);

  return 0;
}
```

```
Hello World!
```

Copy Strings

- To copy the value of one string to another, you can use the strcpy() function:

```c
#include <stdio.h>
#include <string.h>

int main() {
  char str1[20] = "Hello World!";
  char str2[20];

  // Copy str1 to str2
  strcpy(str2, str1);

  // Print str2
  printf("%s", str2);

  return 0;
}
```

```
Hello World!
```

strcpy(copy, what)

strlen

Note that the size of str2 should be large enough to store the copied string (20 in our example).

Compare Strings

- To compare two strings, you can use the strcmp() function.
- It returns 0 if the two strings are equal, otherwise a value that is not 0:

```c
#include <stdio.h>
#include <string.h>

int main() {
  char str1[] = "Hello";
  char str2[] = "Hello";
  char str3[] = "Hi";

  // Compare str1 and str2, and print the result
  printf("%d\n", strcmp(str1, str2));

  // Compare str1 and str3, and print the result
  printf("%d\n", strcmp(str1, str3));

  return 0;
}
```

Strcmp

Strcmp( )

```
0
-4
```

Thank You

len
cat
cpy
$\boxed{\text{cmp}}$

Strlen
cat
cpy
cmp

Ans